[User Help Report: TechSphere]

DATA SCIENCE PROJECT

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1. Project Overview

TechSphere Analysis is a **Streamlit-based intelligent job market explorer** designed to help users search for job opportunities by skills, location, and experience level. It now features:

1- **Interactive job search** with advanced filters

2- **Visualizations** (bar graphs, pie charts, scatter plots, and salary prediction graphs)

3- **Real-time job recommendations via API** (LinkedIn integration)

4- **Live company news feed** related to searched companies

5- **Skill demand analysis dashboard**

6- **User account authentication system**

2. **Key Components:**

File/Notebook	**Purpose**
frontend.py	Streamlit UI for job search, account system, and job cards
backend.py	Data filtering, salary prediction, API integration
Tech_Sphere_Analysis.ipynb	Exploratory Data Analysis (EDA), salary prediction models
Job opportunities.csv	Job dataset
requirements.txt	Python dependencies
README.md	Project documentation
Presentation/	Detailed project PPT

3. Folder Structure

Tech Sphere Analysis/

├── frontend.py

├── backend.py

├── Job opportunities.csv

├── Tech_Sphere_Analysis.ipynb

├── requirements.txt

├── README.md

└── Presentation/

4. Prerequisites

Software:

1- Python 3.8+

2- Libraries: streamlit, pandas, numpy, matplotlib, seaborn, requests, beautifulsoup4

Installation:

pip install -r requirements.txt

Hardware:

1- RAM: 4 GB (minimum)

2- Storage: 500 MB (for dataset and dependencies)

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4. Setup & Execution

1. Clone the repository and navigate to the directory.
2. Ensure **Job opportunities.csv** is in the root folder.
3. Install dependencies:

pip install -r requirements.txt

4. Run the application:

streamlit run frontend2.py or python -m streamlit run frontend2.py

5. Open http://localhost:8501 to access the app.

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5. How to Use the App

Sections:

1- **Home**

2- **Select Job**

3- **Saved Jobs**

4- **Account**

5- **Live News**

6- **Top Skills Dashboard**

Steps:

1. Register/Login to your account.
2. Enter your criteria: (For Database Search)

    1- **Skill** (e.g., "Python")

    2- **Location** (e.g., "London")

    3- **Experience Level** (e.g., "Mid Level")

FOR LIVE JOBS – ENTER ONLY SKILLS (Location is optional)

3. Click **Search Jobs**.
4. View:

    1- **Job cards** with details (title, company, salary).

    2- **Visualizations** (companies, locations, salary trends, demand skills).

    3- **Live company news**.(Apply now)

    4- **Real-time job recommendations**.

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6. Common Errors & Fixes

Error	Cause	Solution
FileNotFoundError	Missing dataset or wrong path	Ensure Job opportunities.csv is in the root folder

Error	Cause	Solution
ModuleNotFoundError	Missing libraries	Run pip install -r requirements.txt
No jobs found	Overly strict filters	Relax search criteria
Salary plot fails	Incorrect salary format in dataset	Format salaries as $50,000-$70,000
News API fetch fails	Network issues or invalid API key	Check internet connection / replace API key if needed

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7. Backend Methods

1- filter_jobs(): Filters job data based on user input.

2- get_skill_demand(): Returns top skills by demand.

3- get_company_news(): Fetches live news for selected companies.

4- fetch_realtime_jobs(): Connects to LinkedIn/Indeed open API for job listings.

5- predict_salary(): Predicts salary based on job description and required skills.

Dataset Requirements:

Columns:

1- Job Title

2- Company

3- Location

4- Required Skills

5- Experience Level

6- Salary Range

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8. Troubleshooting / Edge Cases

1- **Blank visualizations:** Ensure dataset has relevant, non-null values.

2- **News API/LinkedIn API errors:** Verify API keys and network.

3- **Salary prediction errors:** Confirm model dependencies and data formatting.

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9. FAQs

Q: How to add new jobs?
A: Append to Job opportunities.csv in the same format.

Q: Can I deploy this online?
A: Yes! Streamlit Cloud / Heroku supported.

Q: How do I get real-time job updates?
A: The app automatically fetches via integrated APIs — just use the **Real-Time Jobs** tab.

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10. Future Scope

1- Implement **job apply feature**

2- Add **PDF/Excel export**

3- Integrate **company reviews and ratings**

4- Deploy on **Streamlit Cloud with Docker support**

5- Extend **salary prediction model** with job description embeddings

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